Work in progress. AINPI is an experimental exploration of the CMS National Provider Directory (2026-04-09 release). Data may be incomplete, stale, or incorrect — numbers should be independently verified before any business or clinical decision. See provenance analysis → (ainpi.vercel.app/insights)

AINPI · V1.0.0

The State of the National Provider Directory

Six pre-registered findings against the CMS NPD 2026-04-09 release.

NPD release	Methodology	Resources audited	Findings published
2026-04-09	**v0.2.0-draft**	**27.2M**	**6 / 6**

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Executive summary

The CMS National Provider Directory (NPD) ships 27.2M FHIR R4 resources across six resource types. AINPI is an independent, open-source audit of that bulk export. Six pre-registered hypotheses (bundled into six finding slugs) were declared publicly *before* the numbers dropped; this report publishes the results.

The findings cluster in four narratives. First, the NPD's own `meta.lastUpdated` timestamp is a bulk-export stamp, not a per-resource freshness signal — the 30-day and 90-day regulatory update cadences cannot be measured from the bulk files. Second, referential integrity is essentially perfect where references are declared, but coverage of the Endpoint↔Organization link is sparse in both directions. Third, the NPD uses *two different* specialty code systems on two different resources with no cross-walk. Fourth, declared FHIR-REST endpoints clear the 85% Medicare Advantage network-adequacy implied ceiling on basic reachability but not on SMART discovery.

Every number in this report is reproducible from the scripts in the analysis/ (https://github.com/FHIR-IQ/AINPI/tree/main/analysis) directory of the repository. Methodology at

/methodology (https://ainpi.vercel.app/methodology).

H1 · H2 · H3 · H4 · H5

Endpoint liveness

Full crawl of 2,974 distinct FHIR-REST hosts in the NDH: 93.3% answered HTTP, 85.4% served a parseable CapabilityStatement, 81.6% published valid SMART well-known, 90.3% answered an unauthenticated Practitioner?_count=1 with 200/401. Across the full NDH endpoint population: 5,043,524 endpoints total (74.2% FHIR-REST, 25.8% Direct Project); 98.7% of Organizations carry zero Endpoint references.

2.5K / 3.0K = 85.37%

NULL HYPOTHESIS

FHIR-REST endpoints are reachable and conformant at or above the implied 85% network-adequacy ceiling.

DENOMINATOR

All `Endpoint` resources with `connectionType` in the FHIR-REST family.

DATA SOURCE

CMS NPD bulk export + live HTTP probes run by the `ainpi-probe` crawler against declared `Endpoint.address` URLs.

NOTES

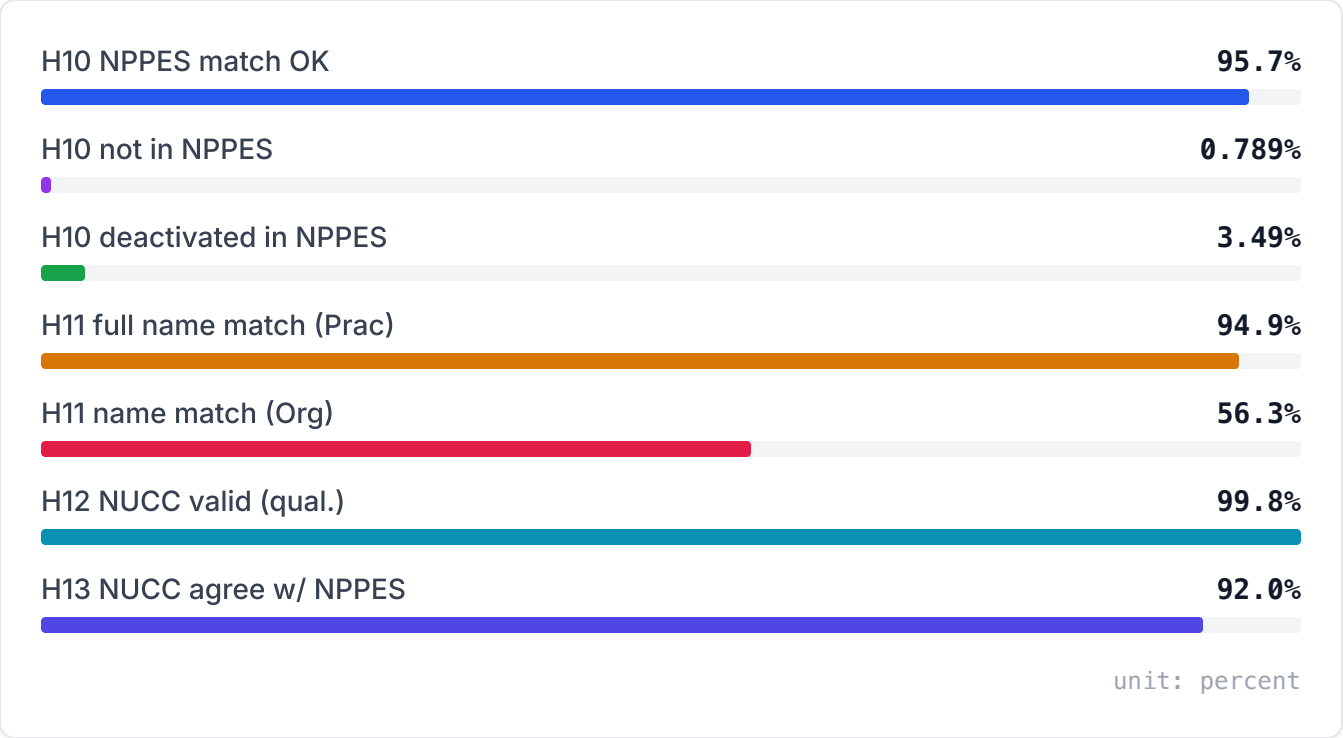
Probed 2,974 distinct FHIR-REST hosts (one endpoint per host, stratified by host). Crawl: 16 global concurrency, 1 rps per host, 10s connect / 30s read, exponential backoff on 429/503, User-Agent AINPI-DirectoryQualityBot/1.0. fhirVersion declared by L5-conformant hosts: {'4.0.1': 1599, '4.0.0': 938, '3.0.2': 1, '3.0.1': 1}. CapabilityStatement software top-5: {'Firely Server': 505, 'Epic': 456, 'Fhir Server': 282, 'Altera FHIR': 71, '1up FHIR Server': 42}. Highest level reached distribution: {7: 2687, 0: 33, -1: 136, 1: 23, 2: 28, 3: 39, 6: 14, 5: 14}. Certs expiring within 90 days of release: 271. H4 (connection type distribution) and H5 (orgs without endpoints) computed over the full population in BigQuery: H4 connection types: {'hl7-fhir-rest': 3741547, 'direct-project': 1301977}; H5 3,603,262 orgs of which 3,556,049 have no managingOrganization reference from any Endpoint.

H9 · H10 · H11 · H12 · H13

NPI and taxonomy correctness

H9 3/10,856,109 NPIs fail structural+Luhn. H10 — 0.789% of 10.9M NDH NPIs are NOT in NPPES; 3.49% are deactivated in NPPES; 0.000% have entity-type mismatch (95.72% clean). H11 — 94.9% of Practitioners match NPPES on full name (exact); 56.3% of Organizations match NPPES on business name (exact). H12 — 99.83% of 7.1M NDH Practitioner.qualification NUCC codes are valid in NUCC v17.0. NDH PractitionerRole.specialty uses CMS Medicare Physician Specialty Types (100.0% of specialty-bearing roles), NOT NUCC. H13 — 92.02% of 7.1M NDH Practitioners agree with NPPES on primary taxonomy code.

`10.4M / 10.9M = 95.72%`



NULL HYPOTHESIS

NPI structural validity is ≥99.9% and NDH-to-NPPES agreement on name and primary specialty is within documented drift thresholds.

DENOMINATOR

All `Practitioner` and `Organization` resources with an NPI identifier.

DATA SOURCE

CMS NPD bulk export joined against the NPPES monthly full dissemination file (V.2) and the current NUCC quarterly code set.

NOTES

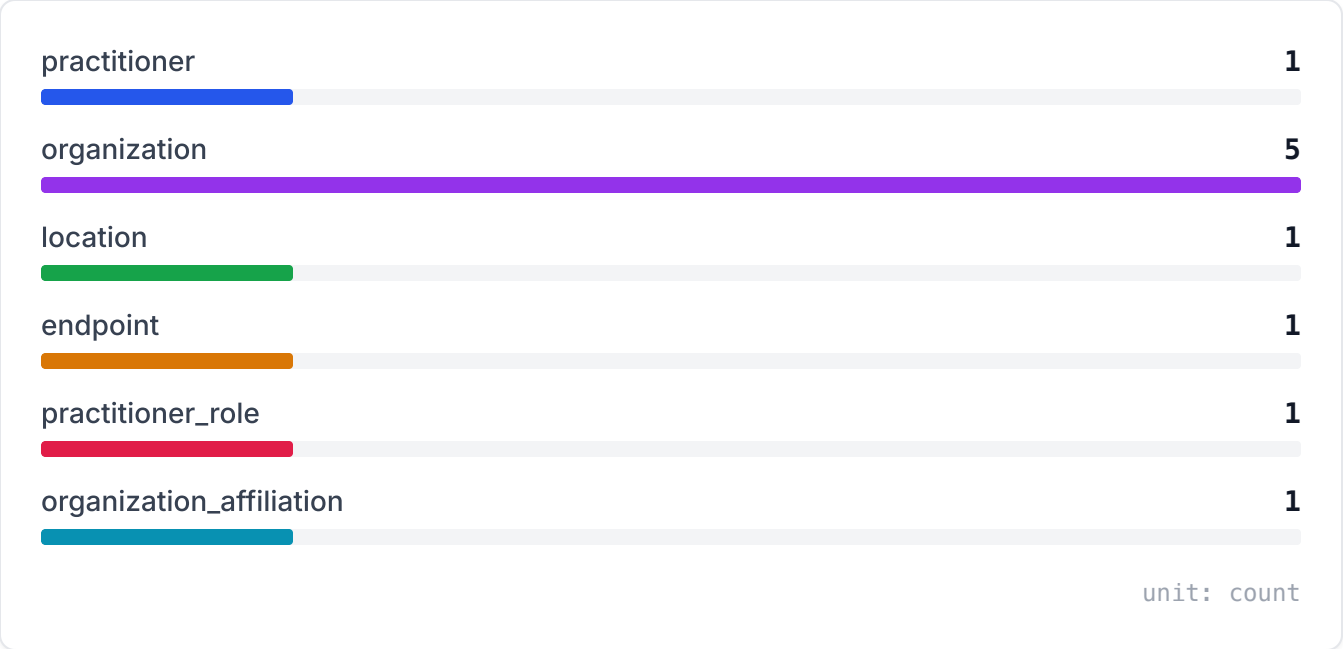
Source: bigquery-public-data.nppes.npi_raw (latest update 2026-02-09, 9.37M NPIs) + .healthcare_provider_taxonomy_code_set_170 joined against NDH 2026-04-09. ~8-week gap window means NPIs deactivated between 2026-02-09 and 2026-04-09 may appear active in NPPES — known caveat. H10: Practitioner — 7,124,478 ok / 56,157 not in NPPES / 260,534 deactivated / 44 enumerated as org in NPPES. Organization — 3,266,647 ok / 29,548 not in NPPES / 118,701 deactivated / 0 enumerated as individual. H11 uses exact case-insensitive match after TRIM+UPPER; Jaro-Winkler ≥0.85 = v2 upgrade. H12 NOTE: NDH PractitionerRole.specialty uses CMS Medicare Physician Specialty Types with system URI https://www.cms.gov/files/document/acceptable-physician-specialty-types-py-2025.pdf (codes like '14-50' = 'PRACTITIONER - NURSE PRACTITIONER'), NOT NUCC. NUCC codes appear on Practitioner.qualification.code with system http://nucc.org/provider-taxonomy (7,124,017 of 7,441,213 Practitioners). H12's metric is validity of those NUCC codes against the v17.0 code set (7,112,042 valid of 7,124,017 populated). NUCC v17.0 is the latest version in bigquery-public-data; if NDH uses a newer code set, codes from later releases would appear invalid here (upgrade to current quarterly NUCC + accept-last-3-years window = v2). H13 compares NUCC primary taxonomy NDH→NPPES only for Practitioners where NDH.qualification.system = http://nucc.org/provider-taxonomy AND NPPES entity_type_code = 1.

H18

Temporal staleness

100.0% of NPD resources carry a meta.lastUpdated value on the release day (2026-04-09). Distinct meta.lastUpdated values range from 1 to 5 across the 6 resource types — meta.lastUpdated on the NPD bulk public-use files is a bulk-export stamp, not a per-resource freshness signal.

`27.2M / 27.2M = 100.00%`



NULL HYPOTHESIS

A majority of resources carry a `meta.lastUpdated` within the 90-day statutory threshold.

DENOMINATOR

All resources across all six NDH resource types that carry a populated `meta.lastUpdated`.

DATA SOURCE

CMS NPD bulk export (pinned release).

NOTES

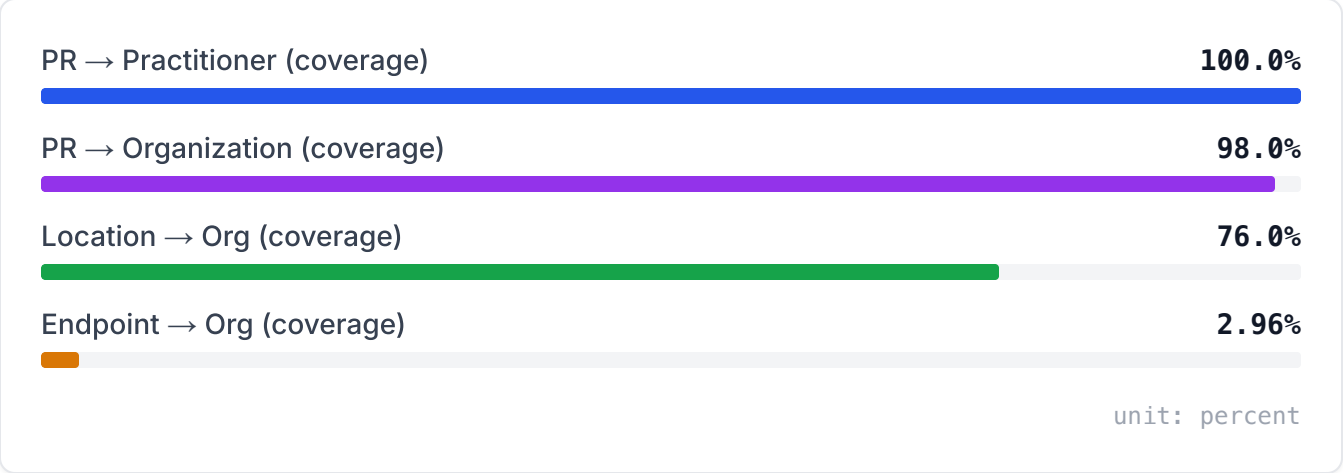
Per-resource distinct meta.lastUpdated values on release day 2026-04-09: practitioner → 1 distinct, 100.00% at modal; organization → 5 distinct, 55.48% at modal; location → 1 distinct, 100.00% at modal; endpoint → 1 distinct, 100.00% at modal; practitioner_role → 1 distinct, 100.00% at modal; organization_affiliation → 1 distinct, 100.00% at modal. Regulatory compliance with the 30-day CMS-9115-F or 90-day REAL Health Providers Act / No Surprises Act update cadence CANNOT be measured from meta.lastUpdated on the bulk files — a per-record freshness signal from upstream NPPES (enumeration_date / last_updated) or PECOS would be required.

H6 · H7 · H8

Referential integrity

Referential integrity is clean but coverage is sparse. 0.000% of 17.0M declared cross-resource references actually dangle (target missing). But only 3.0% of Endpoints carry a managingOrganization (149,080 of 5,043,524) and only 76.0% of Locations do (2,654,922 of 3,494,239). H8: the NPD bulk export does not ship HealthcareService (NDH IG defines 10 resources; NPD ships 6).

0 / 17.0M = 0.00%



NULL HYPOTHESIS

Cross-resource references resolve at ≥99% inside the bulk export.

DENOMINATOR

All reference fields across `PractitionerRole.practitioner`, `PractitionerRole.organization`, `Location.managingOrganization`, and `HealthcareService.providedBy`.

DATA SOURCE

Edge tuples extracted from the NPD bulk export in a single streaming pass, queried in DuckDB.

NOTES

Integrity (dangling rate among declared references): H6a PR→Practitioner 0.0000%, H6b PR→Organization 0.0000%, H7 Location→Org 0.0000%, Endpoint→Org 0.0000%. All near zero — when a reference is declared, it resolves. Coverage (share of rows with the optional reference

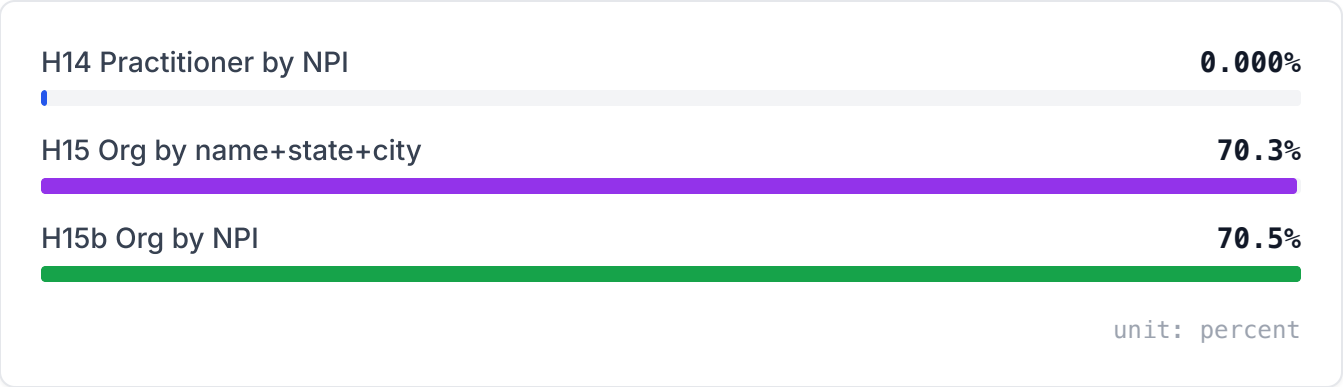
populated): PR→Practitioner 100.00% (required), PR→Organization 97.98%, Location→managingOrganization 75.98%, Endpoint→managingOrganization 2.96%. The Endpoint→Organization gap pairs with H5 (98.69% of Orgs have no Endpoint referencing them) — the Endpoint↔Organization link is sparse in both directions. H8 requires HealthcareService, which is one of four NDH IG resources (HealthcareService, InsurancePlan, Network, Verification) absent from the 2026-04-09 NPD bulk export. Any HealthcareService-based check cannot be performed from NPD alone.

H14 · H15

Duplicate detection

Practitioner dedup is clean — 0 excess rows across 7,441,212 NPIs (H14). But Organizations multiply: 70.5% of the 1,999,118 unique Org NPIs map to more than one Organization resource (1,415,777 excess rows; max 5 resources per one NPI). By normalized (name, state, city), 70.3% of keys repeat. Downstream consumers assuming one Organization resource = one real-world entity will be wrong roughly two out of three times.

`1.2M / 9.2M = 13.35%`



NULL HYPOTHESIS

Duplicate rate is below 1% for both Practitioner (by NPI) and Organization (by normalized name + address).

DENOMINATOR

All `Practitioner` and `Organization` resources.

DATA SOURCE

CMS NPD bulk export.

NOTES

BigQuery dataset has primary-key dedup applied at ingest (-4.6M Practitioner, -383K Organization at _id). These are residual entity-level duplicates. H14 key = _npi on practitioner. Max copies observed: 1 for a single Practitioner NPI. H15 key = (LOWER(name) stripped of LLC/INC/PC/PA/PLLC/CORP/LLP/LTD/CO/COMPANY/THE and non-alphanumerics, UPPER(state),

UPPER(TRIM(city))); orgs with missing name or state or city are excluded. Max copies for one key: 2206. H15-bonus keys by _npi; max copies for one Org NPI: 5. Caveat — some portion of the Organization multiplicity may reflect CMS modeling one FHIR Organization resource per service location rather than true duplication. Either interpretation breaks the common downstream assumption that COUNT(Organization) equals the number of unique organizations. Fuzzy matching (Jaro-Winkler, suite-unit tolerance) is a v2 enhancement.

H22

Network adequacy gauge

Empirical FHIR endpoint liveness vs the 85% Medicare Advantage network-adequacy implied ceiling: L7 unauthenticated-read 90.3% (ABOVE), L5 CapabilityStatement conformance 85.4% (AT), L6 SMART well-known 81.6% (BELOW). Gauge sampled across 2,974 distinct FHIR-REST hosts in the NDH.

2.7K / 3.0K = 90.35%

NULL HYPOTHESIS

Measured endpoint liveness matches or exceeds the 85% regulatory ceiling.

DENOMINATOR

All FHIR-REST endpoints declared in the NPD bulk export at the pinned release.

DATA SOURCE

`ainpi-probe` crawler results joined to the `Endpoint` resource table.

NOTES

The 85% network-adequacy ceiling is the implied minimum active provider share under Medicare Advantage adequacy rules (42 CFR §422.116). This comparison maps 'adequacy' onto technical reachability and conformance — NOT onto the regulatory definition itself, which concerns whether a sufficient share of the network is active, not whether its FHIR endpoints respond. Interpret as: if consumers assume the FHIR directory surface offers a regulatory-equivalent conformance floor, that

assumption holds only on unauthenticated basic reachability (L7 90.3%) and collapses on SMART discovery (81.6% vs 85%). Probe methodology: 2,974 distinct FHIR-REST hosts, one endpoint per host, stratified by host-fingerprint, via ainpi-probe L0-L7 with 1 rps per host rate limit and 10s connect / 30s read timeouts.

About this report

This report covers the AINPI v1.0.0 release. All methodology, pipeline code, BigQuery analyses, and the FHIR endpoint crawler are open source under Apache-2.0:

- Main repo: github.com/FHIR-IQ/AINPI (https://github.com/FHIR-IQ/AINPI)
- Crawler: github.com/FHIR-IQ/ainpi-probe (https://github.com/FHIR-IQ/ainpi-probe)
- Usage examples: github.com/FHIR-IQ/ainpi-examples (https://github.com/FHIR-IQ/ainpi-examples)
- Live site: ainpi.vercel.app (https://ainpi.vercel.app)
- Public URL contract: ainpi.vercel.app/api/v1/stats.json (https://ainpi.vercel.app/api/v1/stats.json)

To cite: see CITATION.cff (https://github.com/FHIR-IQ/AINPI/blob/main/CITATION.cff). To contribute: see CONTRIBUTING.md (https://github.com/FHIR-IQ/AINPI/blob/main/CONTRIBUTING.md). To subscribe: ainpi.vercel.app/subscribe (https://ainpi.vercel.app/subscribe).

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